

James McAllister

Postdoctoral Researcher at Trinity College Dublin & University of Oxford

Theoretical Neuroscience | Applied Mathematics | Neural Computation

james.mcallister@tcd.ie

jajm.uk

LinkedIn

EDUCATION

PhD Theoretical & Computational Neuroscience	2023–2026
Intelligent Systems Research Centre, Ulster University Supervisor: Dr Cian O'Donnell	
Visiting PhD Researcher, University of Bristol (2024) Supervisor: Dr Conor Houghton	
<i>Thesis: "Exploring structure, dynamics, and computation in biologically-inspired recurrent neural networks"</i>	
Associate Fellowship of the Higher Education Authority (AFHEA)	2024–2025
First Steps to Teaching, Doctoral College, Ulster University	
Master of Research (MRes)	2022–2023
Queen's University Belfast Distinction	
PGCE Mathematics	2018–2019
Queen's University Belfast GTCNI Star Award and E. Fulton Prize for Mathematics	
MA (Dubl.) Mathematics	2014–2018
Trinity College Dublin First Class Honours with Gold Medal	

EMPLOYMENT & EXPERIENCE

Postdoctoral Researcher Trinity College Dublin & University of Oxford	2026–present
Computational & Theoretical Neuroscience, Machine Learning, NeuroAI	
Treasurer Institute of Mathematics & its Applications	2025–present
Irish Branch https://ima.org.uk/roles/irish-branch-committee/	
Co-Supervision of Final Year Projects Ulster University	2024–2026
Computational Neuroscience and Machine Learning	
Visiting Researcher University of Bristol	Mar–Apr 2024
Neural Dynamics Group, School of Engineering Mathematics & Technology	
Lecturing & Postgraduate Teaching Assistance Ulster University	2023–2026
Undergraduate & postgraduate tutorials in Mathematics, Algorithms, & Data Science	
Teacher of Mathematics Wellington College Belfast	2019–2022
Mathematics, Further Mathematics, and Physics up to A-Level Further Mathematics, including Complex Numbers, Differential Equations, Group Theory, Vectors & Matrices	

RESEARCH

Geometry of recurrent neural network dynamics for sequence working memory, signal discrimination, and response suppression	
In Preparation: J. McAllister & C. O'Donnell, 2026	
Non-random brain connectome wiring enables robust and efficient neural network function under high sparsity	
J. McAllister , C. Houghton, J. Wade, C. O'Donnell, 2026, bioRxiv doi.org/10.64898/2026.03.30.715411	
Region- & layer-specific glutamatergic synapse development in the nascent cortical hierarchy	
L. Discepolo, J. McAllister , R. Russell, S. Apilado, G. Margetts-Smith, D. Franchini, S.G. Grant, C. O'Donnell, M.C. Ashby, P. Anastasiades, 2026, Journal of Neuroscience doi.org/10.1523/JNEUROSCI.0293-26.2026	

Topological and simplicial features in reservoir computing networks

J. McAllister, J. Wade, C. Houghton, C. O'Donnell, 2024, *Advances in Computational Intelligence Systems* | doi.org/doi.org/10.1007/978-3-031-78857-4_5

Synaptic regularisation of learning via heterosynaptic plasticity

2024–2025

PhD research project

The capacity and accuracy of a triple-well Hopfield model

2023

Group Research Project & Presentation for ISRC Computational Neuroscience Autumn School

A discrete attractor model of decision making

2023

Group Research Project & Presentation for Neuromatch Academy Summer School

A review & statistical analysis of the effect of formative assessment in mathematics

2023

Research Project & Dissertation, Distinction, Queen's University Belfast

A multilevel analysis of high-stakes examination results in mathematics

2021

I. Cantley & J. McAllister | doi.org/10.1007/s11199-021-01234-5

Trigonometric series and the emergence of transfinite set theory

2018

Final Year Research Dissertation & Poster, First Class (Distinction), Trinity College Dublin

SKILLS & INTERESTS

Languages : English, German, French, British Sign Language

Programming : Julia (main), Python, MATLAB, R, SPSS

Tools : High Performance Computing, L^AT_EX, Microsoft, Google Suite

Interests : Theoretical neuroscience, Mathematical modelling, Network theory, Dynamical systems, Complex systems, Mathematics of Machine Learning, Assessment theory

TALKS & POSTERS

Talk : *All Ireland Computational Neuroscience Symposium*, TU Dublin Aug 2026

Talk : *University of Oxford*, Centre for Integrative Neuroimaging Jul 2026

Poster : *Research in Encoding & Decoding of Neural Ensembles (AREADNE)*, Milos, Greece Jun 2026

Poster : *Intelligent Systems Research Centre Discovery Day*, Ulster University Jun 2026

Talk : *Festival of Research*, Ulster University Jun 2026

Talk : *All Ireland Computational Neuroscience Symposium*, Queen's University Belfast Dec 2025

Talk : *NI High Performance Computing Conference*, QUB [link] Nov 2025

Talk : *Dynamical Systems and Neuroscience*, Nanyang University [link] Oct 2025

Talk : *Recent Trends in Rough Analysis & Dynamical Systems*, Liverpool University [link] Jul 2025

Poster : *UK Neural Computation*, Imperial College London Jul 2025

Seminar : *Ignite Sessions*, Intelligent Systems Research Centre Jun 2025

Talk : *University of Cambridge*, CBL Lab [link] Jun 2025

Poster : *Festival of Research*, Ulster University Jun 2025

Poster : *CoSyNe*, Montreal & Mont-Tremblant [link] Mar 2025

Talk : *Intelligent Systems Research Centre*, COIN Club Feb 2025

Talk : *Workshop UK Computational Intelligence*, UKCI 2024 Sep 2024

Talk : *Computational Neuroscience, Neurotechnology & NeuroAI Summer School* Aug 2024

Poster : *UK Neural Computation*, Sheffield [link] Jul 2024

Poster : *Conference on Mathematical Neuroscience*, Dublin [link] Jun 2024

Seminar : *Intelligent Systems Research Centre* May 2024

Talk : *Intelligent Systems Research Centre*, COIN Club May 2024

Talk : *University of Cambridge*, British Society for Research into Learning Mathematics Mar 2020

ACHIEVEMENTS & AWARDS

Best Poster Prize, ISRC Discovery Day, Ulster University	Jun 2026
Best Poster Prize, PhD Festival of Research, Ulster University	Jun 2025
Travel Grant, Computational and Systems Neuroscience (CoSyNe)	Mar 2025
Best Student Paper Award, UK Computational Intelligence, Belfast	Sep 2024
Visiting Scholarship, University of Bristol	Feb 2024
PhD Scholarship, NI Department for the Economy	2023–2026
E. Fulton Prize for Mathematics, QUB	Sep 2019
GTCNI Star Award, PGCE, QUB	Jul 2019
Gold Medal, Trinity College Dublin	Nov 2018
Trinity College Dublin First Class Prize	2015, 2016, 2017
Naughton Scholarship	2014–2018
Exhibition Award, Trinity College Dublin	Sep 2014
Trinity College Dublin Sizarship	2014–2018

COURSES & TRAINING

Open Brain Institute: Spines & Motifs Participation Webinar	Apr 2026
Introduction to Neuronal Modeling (DendroTweaks & Dendrifry), Poirazi Lab	Apr 2026
Fundamentals of Accelerated Computing with Modern CUDA C/C++, NVIDIA & NI-HPC	Nov 2025
Hausdorff Centre: Statistical mechanics of spin glasses, neural networks & learning	Jun 2025
Associate Fellow of the Higher Education Authority (AFHEA), First Steps to Teaching	2024–2025
Training Sessions for the Northern Ireland High Performance Computing system	2023–2025
Computational Neuroscience Autumn School, ISRC, Ulster University	2023, 2024
Computational Neuroscience Neuromatch Academy Summer School	Jul 2023
INCF Python-based modelling course	Mar 2023
British Sign Language Level 1	2019–2020

ACADEMIC & PROFESSIONAL SERVICE

Session Chair: Ulster University PhD Festival of Research	2026
Peer Reviewer: PLOS Computational Biology, Neural Networks	2025–
Treasurer: Irish Branch, Institute of Mathematics and its Applications	2025–
Volunteering: NI Maths Fest; Organising International Women in Maths Day	2025, 2026
Joint Acting Head of Department (Mathematics): Wellington College Belfast	2021–2022
Numeracy & Charity Coordinator: Wellington College Belfast	2021–2022
Speaking: Mathematics PGCE Events, Queen's University Belfast	2019, 2020
Assistant: BSRLM (QUB) Open Days, School of Mathematics, TCD	2015–2019

REFEREES

Dr Cian O'Donnell

Research Director & Senior Lecturer in Computational Neuroscience and Data Analytics, Ulster University
c.odonnell2@ulster.ac.uk

Dr John Wade

Lecturer in Electrical and Mechanical Engineering, Atlantic Technological University
john.wade@atu.ie

Dr Conor Houghton

Reader in Mathematical Neuroscience, University of Bristol
conor.houghton@bristol.ac.uk